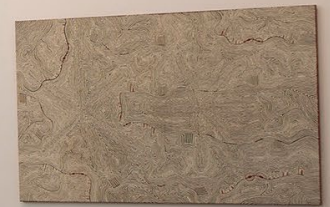


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INCLUDED INSIDE:

Fire Alarm System and Elevator Integration for Occupant Evacuation Operation
"I Don't Know What Those BACnet Web Services Are...But I Think I Want Them."
Addressing IP Security Concerns when Deploying a BACnet System
ASHRAE SSPC 135 Meetings BACnet Proceedings
Have a Question About BACnet? The BACnet Institute has your Answer!

Overview

The Foundations publication is an educational resource produced by BACnet International for its members. BACnet International is the cornerstone of your success, and Foundations builds on that by providing the ground level knowledge in connecting the dots in building automation. Foundations is written by volunteers from the BACnet community for integrators, installers, appliers and specifiers/consultants. It complements the BACnet International Journal and the association's monthly enewsletter, Cornerstones.

For more information on BACnet International, please visit www.bacnetinternational.org.

Questions or article submissions may be directed to BACnet International Association Manager Natalie Nardone, at natalie@bacnetinternational.org



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The Australian National University
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Fire Alarm System and Elevator Integration for Occupant Evacuation Operation

By **Axel Kunze**, Head Software Architect, Fire Panels Development, Siemens; **Andy Huber**, Senior Key Expert, Fire Detection Domain, Siemens; **Christopher Mason**, Senior Manager - Internet of Elevator and Escalator Development, Schindler Elevator Corp.

Background

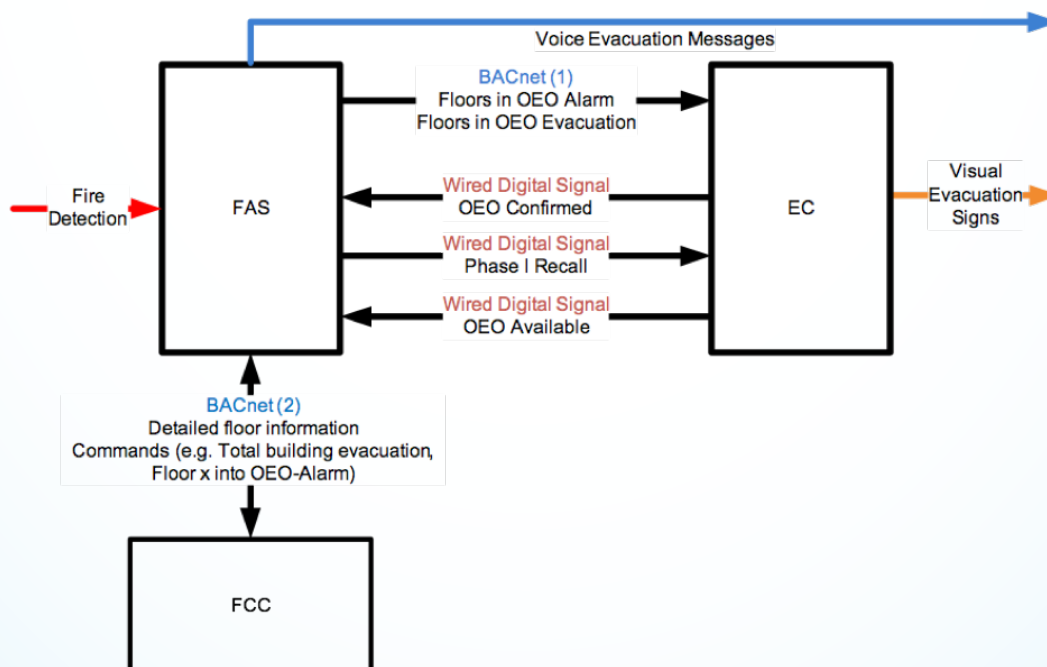
The evolution of the building industry and elevator technology in the wake of the Sept. 11, 2001 attacks has been strongly influenced by the need to safely and efficiently evacuate mid- to high-rise buildings in the event of an emergency. Building, fire alarm and elevator codes have been updated to include new evacuation functions and interfaces for the use of elevators for evacuation even in case of a fire in the building, referred to as “Occupant Evacuation Operation” (OEO). The recently released ASME A17.1 2013 code lays out requirements for the evacuation of building occupants under conditions where a fire remote from the elevators is detected. This is the first time elevators are an essential part of the evacuation strategies of a building under such circumstances.

OEO requires the interoperation of different building systems: the fire alarm system including the voice evacuation system, and the elevator system. This makes it obvious that a tight and robust information exchange between the fire and elevator systems is crucial for a successful realization of OEO.

The straightforward approach to implement a robust communication between fire alarm systems (FAS) and elevator system (ES) is via individual, hard-wired binary signals per floor. However, this involves a high wiring and engineering effort and cannot be considered state-of-the art. We show that BACnet is well suited to cover OEO.

System Overview

The following figure provides an overview of the integration of the FAS with the ES for OEO over BACnet.



The first logical BACnet connection (labeled BACnet(1)) between the FAS and the ES transports the floor states to the ES. This includes the floors that are on fire (i.e. in Alarm state) and the adjacent floors that need to be evacuated also (i.e. in Evacuation state). For separation of regulation domains, this is strictly a one-way communication with the ES being the client of information provided by the FAS as a server. No commanding of the FAS from the ES is foreseen here.

The second logical BACnet connection (labeled BACnet(2)) is the one between the FAS and the fire command center (FCC). Again, the information is provided by the FAS on a per-floor basis.

As an addition to the BACnet communication between the FAS and the ES, for compliance with standards and codes, three distinct wired digital signals are used, to ensure the required safe backup behavior in case the BACnet connection is disturbed.

The FAS is responsible for providing information about the location of a fire and the floors to be evacuated and managing the voice evacuation. For OEO, it is relevant on which floor the fire has been detected. Moreover, alarms that prevent OEO must be identified (e.g. a fire in an elevator lobby or in the elevator's machine room).

The ES is responsible for handling elevator car calls according to the fire propagation and the floors to be evacuated as reported by the FAS. The ES also controls the visual evacuation signs to provide guidance to the occupants in the elevator lobbies and while traveling in the elevator.

A fire command center (FCC) typically provides a graphical representation of the building and the floors that are on fire and the floors to be evacuated. The FCC allows manual intervention, e.g., manual evacuation of individual floors, or initiating a total building evacuation.

One of the most important requirements is that the voice evacuation messages which are controlled by the FAS, and the visual evacuation signs controlled by the ES, must always be consistent.

When either the FAS or the ES notices that its counterpart is missing, no OEO is possible. When using a communication protocol like BACnet, continuous supervision of the communication and presence of the counterpart is required.

Modeling Floors

The information that OEO deals with is connected to floors (aka building levels), and elevator groups. Therefore the floor has been chosen as the core element of the OEO domain model. Floors are typically identified by their universal floor number (UFN), a numbering scheme starting with 1 for the absolute lowest floor of a building, and incremented for each higher floor.

In large buildings, elevators are usually organized in groups. Therefore the concept of elevator groups is also presented in the model. Essentially, the elevator groups have a structuring function. Some states, like an elevator's availability for OEO, are always synchronized throughout an entire elevator group.

Floor representations by the FAS contain the main part of the OEO logic to be handled by the FAS. Each floor has its state that is evaluated and published by the FAS. The ES uses these states to determine the priorities with which the different floors need to be served.

Logically, the following states of the floor representations in an FAS are essential for OEO and the ES:

- Alarm** The floor is on fire.
- Evacuation** The floor is one or two floors above or below a floor on fire, or the floor is between two floors on fire.
- Not Served** The floor is not served because of an active evacuation of other floors.
- Unavailable** Elevator's OEO is not available for this floor.
- Phase 1 Recall** At elevator group level, all elevators of the group are recalled to the discharge level (normally the exit floor). No OEO is possible.

BACnet Model

The FAS models each floor to represent its state. As the OEO is evaluated individually per elevator group, the fire system has a group representation called OEO group that contains the OEO floors. The OEO floors and the OEO groups both have two main properties: First, the process state indicated which includes OEO states evaluated by the FAS and second, the operation mode of OEO floors and groups that can be controlled from outside.

BACnet does not specify how OEO floors and OEO groups have to be represented in BACnet. Some FAS vendors are effectively using the standard life safety objects (Life Safety Zone) to represent the OEO floors and groups. Other vendors may have chosen to use Multi-state objects (Multi-state Input, Multi-state Output, and Multi-state Value objects) for representing this. And, very basic, it might be possible to represent OEO relevant states as simple binary signals, through Binary objects (Binary Input, Binary Output, and Binary Value objects).

In the future, standardized semantic tags on objects will help identify those objects that provide the OEO floor and OEO group representations, process state indication, and operation mode control.

FAS Representation for OEO by Life Safety Objects

The FAS may use the standard life safety objects to represent OEO floors and groups. Both can be represented by Life Safety Zone objects that contain a `Present_Value` and a `Mode` property.

The `Present_Value` of the Life Safety Zone object is used to indicate the current state of the OEO floor or OEO group. At this point, no standardized states are available in BACnet for OEO, but proposals for standardization are on the way in the BACnet committee.

The `Mode` property represents the current operation mode of the OEO floor or OEO group, and enables to control the OEO floor or OEO group by writing desired operation modes to this property. A list of all supported modes is present in a third property named `Accepted_Modes` which is a list of values from the `BACnetLifeSafetyMode` enumeration.

FAS Representation for OEO by Multi-state Objects

As an alternative, an FAS may use BACnet Multi-state Input, Multi-state Output or Multi-state Value objects for representing the state and mode of the OEO floors and OEO groups. In Multi-state objects, the states indicated by the `Present_Value` property is vendor defined entirely. For operation modes, dedicated Multi-state objects are typically used, and the desired mode is written to the `Present_Value` of the Multi-state object that represents the mode of the OEO floor or OEO group. As for the states, the operation modes are entirely vendor defined.

FAS Representation for OEO by Binary Objects

Also, one could think of using BACnet Binary Input, Binary Output or Binary Value objects to represent the states, etc. This would be a very limited representation requiring a larger number of objects, and therefore unlikely to be seen.

BACnet Life Safety Object Model for OEO

Fire panels of an FAS may provide a BACnet interface for both FCC workstations and other clients, but also for integration with other building automation domains. On this interface, the fire domain, including evacuation control by voice messaging, can be represented using the BACnet life safety objects. For integration with an ES for OEO, it is just a natural evolution of such fire panels to extend this BACnet interface and the life safety objects to support OEO floor and OEO group representations, including specific states and modes for OEO.

The authors would like to thank Bernhard Isler for his contributions and consultancy to this project and this article.

ABOUT THE AUTHORS

Axel Kunze has been working in software development for 23 years, 10 years as a software architect with a special focus on networking. He heads the software architects team for Siemens fire panels development in Zug, Switzerland. Alex is a co-author of the white paper on BACnet integration of elevators and fire systems for OEO.

Andy Huber has been working in the development of Fire Alarm Systems for 24 years, mainly as a requirements engineer. He is a Senior Key Expert in the Fire Detection Domain at Siemens in Zug, Switzerland. Andy is a co-author of the white paper on BACnet integration of elevators and fire systems for OEO.

Christopher Mason has been working in the development of elevator controllers and advanced remote diagnostic platforms for 30 years. He is the Senior Manager – Internet of Elevator and Escalator Development at Schindler Elevator Corp. in Morristown, NJ. He is a co-author of the white paper on BACnet integration of elevators and fire systems for OEO.

“I Don’t Know What Those BACnet Web Services Are... But I Think I Want Them.”

By Nicolas Waern, Head of Strategy & Business Development, Go-IoT;
Throstur Jonsson, CEO and Founder, Go-IoT

Is BACnet/WS Changing BACnet to IoT net?

It is a common misunderstanding that BACnet only has to do with Building Automation. This is simply not the case, because BACnet is a full-blown device-to-device communication standard. Most importantly, it defines the Application layer of the 7 layer ISO model. This is very important as it gives devices from different vendors the capability of communicating intelligently with each other right from installation. BACnet is a globally recognized standard which is adapting well to new technologies and market needs in the Internet of Things (IoT) sphere. A good example is the introduction of the BACnet/WS (WS = Web Services) and lately the BACnet/IT, about which Bernard Isler from Siemens wrote a very interesting article in the last issue of the BACnet International Journal.

One of the main benefits of BACnet/WS is that it basically encapsulates the BACnet network in a well-behaved and structured RESTful API. This allows for much easier interoperability between building automation systems (BAS), enterprise applications and anything going to the cloud. In short, BACnet/WS is able to create an ecosystem without the worry of huge switching costs and vendor lock-in effects.

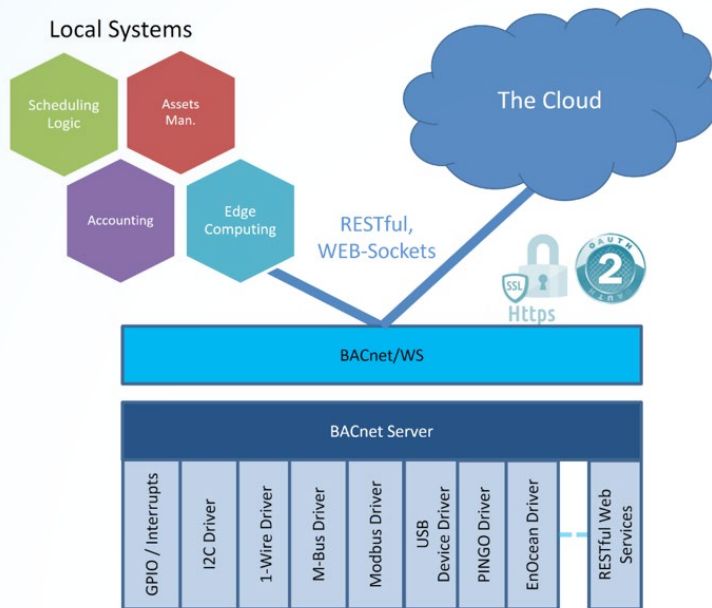
With the age of IoT, the ‘direct-to-the-cloud’ strategy might have to change soon enough due to the massive amounts of data. There will be a need for other solutions closer to the origin, where the data is created, which we will touch upon in this article. Smart homes and smart buildings will become the new norm, and cognitive self-learning buildings are not necessarily that far ahead on the horizon.

Enable New, Secure, Business Opportunities Under One Umbrella

Perhaps the principal obstacles towards a true eco-system are the technological legacy systems, which are hindering innovative and effective technologies. Most industry experts agree that if we enter the realms of building automation, we will find that most buildings are not ready for the future, or even the present. They have outdated technology that lacks the needed security, and require serious engineering effort in order to get to where they need to be. Also, if they have a setup from different vendors, documentation can often be severely lacking. All of this combined makes for a very challenging shift to the world of cognitive buildings and smart cities.

There is a need for improved security and safety, space optimization, energy and lighting efficiency and, of course, improving the health and wellbeing of tenants. However, prior to making sense of Big Data, one must have access

to the data with the ultimate purpose of controlling that data for a desired outcome. BACnet can be the necessary umbrella that harmonizes the different objects in the building, allowing device-to-device communication as previously mentioned - and that's not the end of it. BACnet/WS opens exciting new business possibilities for building owners and others to be able to monetize in new and revolutionary ways. We are certain that this is where BACnet and BACnet/WS will play an even bigger role in the future.



Picture 1.
Go-IoT solution of encapsulating "everything" under BACnet, and open it up to the cloud and local systems using BACnet/WS

Those of us in charge of the BACnet standard decided from the beginning that we would implement the Web Services part (that is BACnet/WS) even before that part of the standard was fully defined. This was because of the need to connect IoT devices to the Internet securely and in a standardized manner without losing functionality. All communication, whether it is internal or external authorization, runs over HTTPS - that is a minimum requirement. BACnet/WS is very comprehensive, making it possible to perform whatever operation is needed over the Internet. This is not true for other simpler standards like MQTT. Standards like MQTT can, therefore, easily be implemented to perform a subset of BACnet/WS, particularly for mobile connections where connection loss may be expected for short or long periods.

Enabler of Truly Cognitive Buildings

BACnet/WS is a vital building block in enabling tenants to talk to their building in a natural way, and for the building to talk back. For such a cognitive purpose Go-IoT has implemented full support for communicating with IBM® Watson — a cognitive cloud-system.



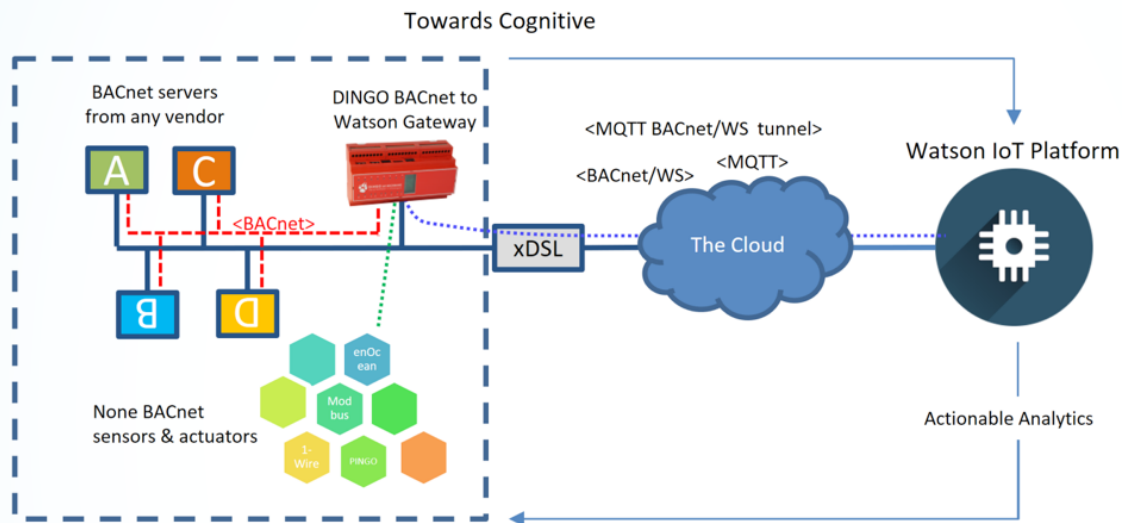
Picture 2. Go-IoT laboratory dashboard via BACnet/WS to IBM Watson gateway.

Implemented in Node-RED and hosted in the IBM-Watson cloud.

The implementation communicates via pure BACnet/WS or via IBM's MQTT-based device-to-cloud interface. As said before, MQTT as such can only handle a subset of BACnet/WS functionality. Therefore, we also implemented a so-called BACnet/WS-tunneling via MQTT to eliminate those MQTT-limitations. This allows the data users to:

- proactively learn from shared data across buildings in the portfolio,
- gain insights and adopt best practices of other buildings in the ecosystem,
- discover trends and analytics based on theirs, and 3rd party data and services, and
- return cognitive commands from IBM Watson to devices in the building.

By gaining access to the IBM Watson and IBM Bluemix services, the owner of the data would also be able to explore possibilities with voice recognition, face detection and, in the not so distant future, a dialogue between humans and buildings. This will not be possible if data flows back and forth in tendrils only, which makes it important that BACnet/WS opens the floodgates of data between the buildings and the cloud. This might involve new BACnet standard objects like 'Camera'. The new BACnet/IT previously mentioned will be an important module in this context.



Picture 3.
Harmonizing
data from any
vendor and send
it to the cloud
and back

BACnet/IoT is our implementation of the BACnet, BACnet/WS, BACnet/AM (App Manager) and BACnet/WS BACnet-builder, a cloud based, device and gateway configuration utility. This is our desire - to extend BACnet to the world of IoT, thus gaining the more mainstream attention it deserves.

Get the Ability to Mix Anything and Everything

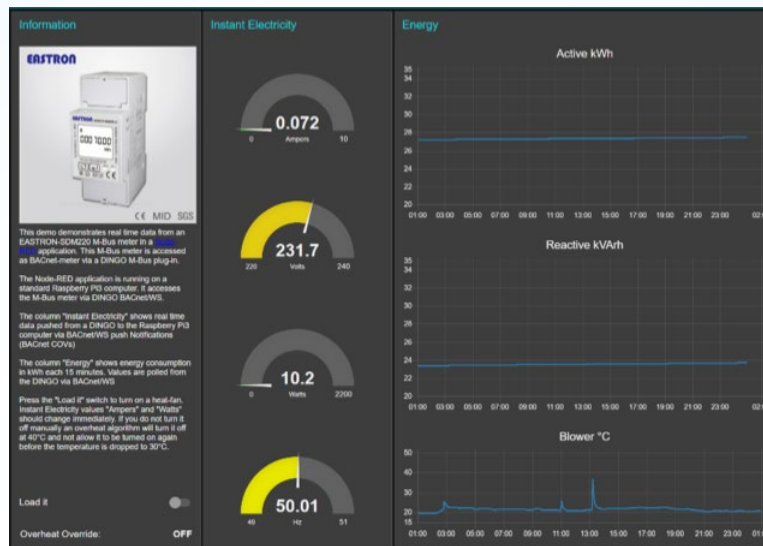
Engaging in a dialogue with customers is always a learning experience, where new exciting use cases develop through brainstorming sessions. As soon as we introduced BACnet/WS to our first POC clients, we instantly got questions about potential use cases. These included:

Q. "Why can't BACnet devices that communicate directly with web-services on the cloud be seen as BACnet objects? For example, can the A/C & heating system gather outside temperature from weather forecast services?"

A. In order to address this, we started implementing standard BACnet Analog Input Objects where the present-value property is coming from a weather forecast temperature in the cloud. The result was that even a standard BACnet trend log object was able to trend log the future.

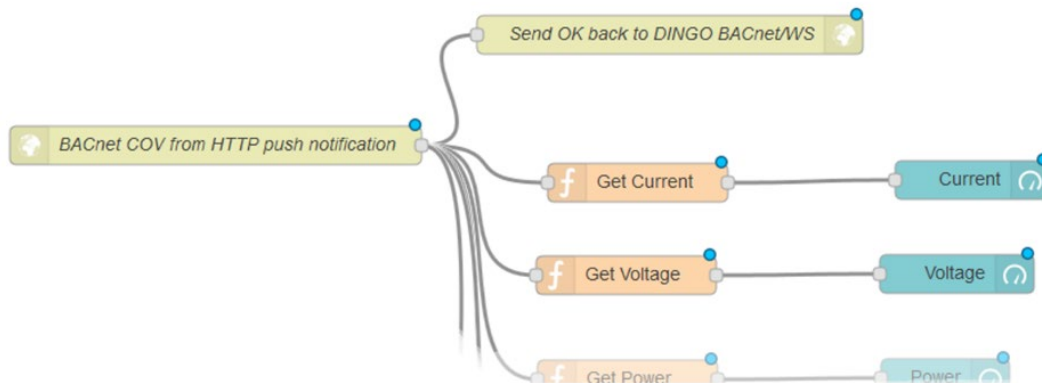
Q. "Can I use WYSIWYG (What You See Is What You Get) tools like Node-RED to create a business logic application that seamlessly communicates with the BACnet?"

A. Yes, absolutely. BACnet/WS is REST-based and therefore integrates perfectly with JavaScript-based tools like Node-RED that is becoming a de facto standard in IoT applications. Additionally, one can run the Node-RED locally on a building device, such as our DINGOs or Raspberry Pi, a PC, and in the cloud. The location of the application does not matter, thanks to BACnet/WS. We demonstrate this in two demos where both applications run against the same BACnet network in our laboratories.



Picture 4. A simple Node-RED dashboard using BACnet/WS

We strongly believe that RESTful Web Services are becoming the glue that interfaces “anything to everything.” Those are on top of BACnet, but can also be under BACnet. BACnet/WS is therefore a vital component of migrating BACnet to whatever can be imagined in terms of applications and cloud services, seamlessly and transparently, without losing any BACnet functionality. The introduction of the cutting edge BACnet/IT addendum makes this picture even more homogenous.



Picture 5. Programming made simple with Node-RED

Once everything is connected an impressive overview is gained, where different technologies are hidden beneath the BACnet umbrella and only the desired objects appear which can be visualized in a dashboard. Once this has been implemented, a decision can be made from a business perspective about what should be done to meet the future in the most effective way. It is then possible to open the floodgates, to connect seamlessly various local enterprise applications, or to the web of IoT (WoT). It doesn't matter anymore where the individual items of the total solution are located, thanks to BACnet/WS. However, as stated in the opening paragraph, straight to cloud is not always the best solution.

Perfectly Positioned to Take the Lead as the IoT Standard

BACnet as a standard is evolving fast. The recent addendum for BACnet/IT and the establishment of the BACnet Application Profile Working Group, preparing a dictionary of tags to give BACnet objects semantic meaning, is proof that BACnet is fast becoming a powerful IoT standard. Another popular project called “Project Haystack” addresses this dictionary of tags issue. Will that project merge with the BACnet definitions or vice versa? Only time will tell but hopefully we will not see two standards within that context.

BACnet/WS will have a very important role to play in a data driven world. It not only acts as the bridge between web services and BACnet, but it will also be the secure link between the automation realm and the cloud, integrating the current isolated BACnet islands seamlessly into the cloud of cognitive services and analytics. Wireless low-power sensors are emerging in the market right now with a battery-life of over 10 years. Some of those even harvest their energy from the surrounding environment.

The simple installation procedure of such sensors makes them very cost effective, and will greatly increase the number of installed sensors and actuators. This will further require Edge Computing strategies and a clear definition of what data stays on site and what goes to the cloud. Standardized protocols like BACnet and BACnet/WS will help a lot in making that happen.

A short explanation of Edge Computing is when data must be processed near the source and only some overview data will be sent back to the cloud.

A practical example is an industrial PID-regulation loop implemented in Node-RED and running in a Backbone hardware box, a kind of low cost Edge Computing Server. The PID loop processes real-time data fast, and must stay close to the controlling hardware. Results from the PID-control can be sent to the cloud; subsequently the PID-loop can receive its control parameters back from the cloud after some cognitive computing or human intervention.

In this practical example, we can also utilize the waste of CPU power and memory inside our hardware box to run the interface to the cloud (BACnet/WS), manage authentication (OAuth 2.0) and encapsulate non-BACnet devices like EnOcean sensors or M-Bus meters (via plugins), under the BACnet umbrella. As such, those non-BACnet devices show up as any other BACnet device.

The possibilities are endless considering the fact that BACnet can be used in other settings in addition to the world of building automation.

Harness the Great Possibilities with BACnet/WS

The intent with this article was to provide some information about BACnet/WS for building owners, integrators and BACnet aficionados so they could go out and explore the possibilities of connecting worlds of data that are ready for the taking. Because even if we believe that data will rule the world, we also understand that a well-designed standard that integrates BACnet seamlessly and securely into the cloud of Web Services and Edge Computing will become extremely important for the future. The ones who harness the power of BACnet/WS are the ones who will have a head start, innovating the space of building automation and beyond. We are confident that BACnet/WS is positioned to become the de facto standard of the future, and it is our sincere recommendation that everyone get started with BACnet/WS and create the change you want to see in the world of IoT.

For more information, please contact Nicolas Waern at nicolas@go-iot.io or Throstur Jonsson at tj62@go-iot.io from Go-IoT.

ABOUT THE AUTHORS

Nicolas Waern is the Head of Strategy and Business Development at Go-IoT. He has an academic background in International Management and IT Management, and extensive experience with being the bridge between business and IT.

Throstur Jonsson is CEO and founder of Go-IoT. He received a MSc. in Electric and Software Engineering from Denmark University, and has long-term experience in technical software development and device data communication. Throstur recently joined the BACnet Data Modeling Working Group.

Addressing IP Security Concerns when Deploying a BACnet System

By **Harpatap Parmar**, Senior Product Manager, Contemporary Controls

BACnet is now the de-facto standard for building automation and management. It is an open protocol, works across devices from multiple vendors and its popularity will only further increase with time because of these factors. As with any protocol that becomes popular, the “Security by Obscurity” approach is not a valid approach. Almost everyone has heard about HTTPS or Secure HTTP. This security is ensured with the use of IP security protocol called Transport Layer Security (TLS). TLS is an industry standard being used for securing millions of HTTP transactions over the internet. Currently, TLS version 1.2 is in use. Its predecessor was Secure Sockets Layer (SSL). SSL version 1.0 (SSLv1) was developed by Netscape but was never publicly released. Version 2.0 and 3.0 were developed in the mid-1990s and used until a few years ago. SSL version 3.0 was the basis for TLS version 1.0 but SSL and TLS were not interoperable. As security flaws were discovered in SSL, the use of SSLv2 and SSLv3 was deprecated. TLS version 1.0 and 1.1 also suffered the same fate and only TLS version 1.2 is the recommended version that should be used to provide security. Even though SSL is no longer in use and has been replaced by TLS, the term “SSL” continues to be used to refer to TLS. BACnet has its own Network Security definition which allows standard BACnet messages to become secure. BACnet Web services, BACnet/WS, can implement TLS and OAuth to provide security. OAuth (Open Authorization) is an open protocol that allows secure authentication and authorization. This is used to grant access to data for authorized applications. BACnet/WS which can utilize HTTPS and TLS is a more industry standard way of communicating in a secure manner. But not all BACnet devices support BACnet/WS. Furthermore, there are many devices where it is not possible to implement TLS or BACnet/WS. It is also not feasible to update all the legacy systems and applications in use to add security without incurring a high cost. Other options need to be used in such cases to fulfill the security needs when utilizing BACnet.

Defined	
Protocol	Year
SSL 1.0	n/a
SSL 2.0	1995
SSL 3.0	1996
TLS 1.0	1999
TLS 1.1	2006
TLS 1.2	2008
TLS 1.3	TBD

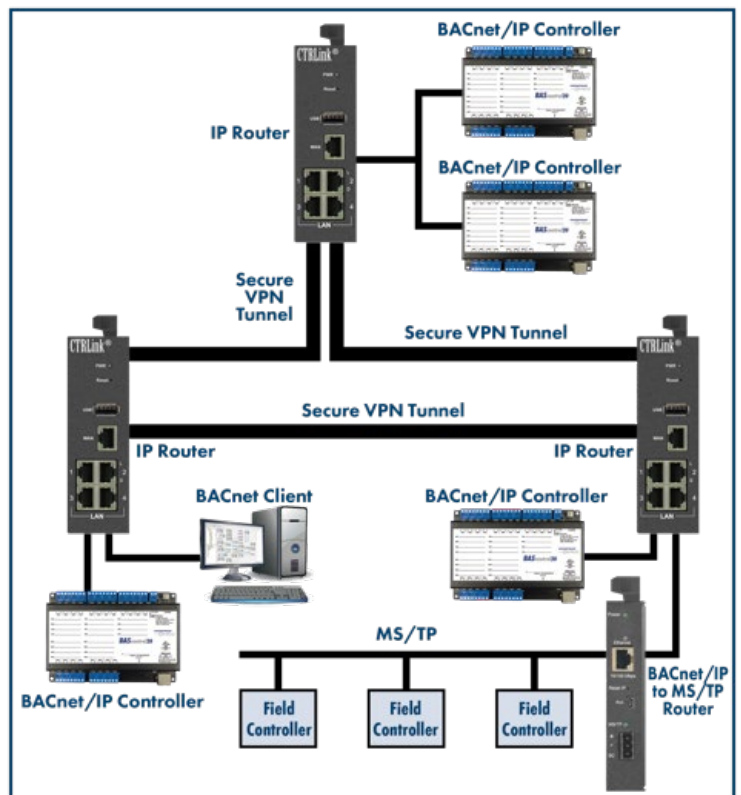
TLS Security

TLS or Transport Layer Security is the widely-used protocol to safeguard the internet and its communication by providing encryption of data. TLS provides confidentiality with the use of encryption. The encrypted data can be freely transmitted over the open internet without the ability to decipher the original data from it by a third party. Encryption involves the use of keys where the sender encrypts the data using a key and the receiver decrypts the data using a key. If the same key is used by both parties, it is called Symmetric Encryption. If different keys are used for encryption and decryption, then it is called Asymmetric Encryption or Public Key Encryption. Data integrity is provided with the use of hashing algorithms that safeguard against data tampering or loss of data due to communication problems. Hashing is a similar concept like having checksums but the hash algorithms are more sophisticated where the change of any data will result in a different hash. Two different sets of data should never produce the same hash and if it is found to be the case, the use of that hashing algorithm is then discouraged. A larger hash calculated on the data sets will be different and is the reason why the hash lengths in use keep increasing. TLS also provides authentication by ensuring the identity of the devices with the use of digital certificates. The use of certificates ensures that only authorized devices can communicate with each other.

VPN as an Alternative Approach

For devices that don't implement BACnet/WS or devices with minimal memory where it is not possible to implement BACnet/WS with TLS or BACnet Network Security, a separate device like an IP router implementing VPN can provide security. VPN or Virtual Private Networks involve creating a secure connection between devices/networks. This secure

connection is generally called a VPN tunnel and provides a way to send encrypted data. VPNs also use the TLS protocol to implement security. The data encryption takes place at the edge and the end devices are not computationally burdened with the task of encrypting or decrypting traffic and are free to perform their BACnet functions. Also, the existing applications don't need to change to add BACnet/WS to provide security. The BACnet device on one end sends the traffic where it is encrypted by one IP router and the encrypted data is then sent over the VPN tunnel to the IP router on the other end. The second router receives the encrypted data, decrypts it and the normal BACnet data is forwarded to the recipient BACnet device. Any type of data can be sent over the VPN and that includes BACnet/IP traffic. All the BACnet traffic utilizing any of the standard UDP ports, 0xBAC0 to 0xBACF, can be easily sent over this VPN tunnel and there is no need to make changes to the firewall configuration to accommodate the various BACnet UDP ports. The only port that needs to be configured in the firewall is the VPN port. VPN also has the added advantage that the end devices and the infrastructure behind the IP router providing VPN are hidden. It can be used to securely connect and monitor remote sites and program BACnet controllers. BACnet/MSTP devices can be used in conjunction with a BACnet/MSTP to BACnet/IP router and can also be communicated to securely over the VPN. VPNs are flexible and can be used to connect two sites or multiple sites securely. This is useful if it is a large campus and multiple buildings need to be connected. VPNs also provide the ability to configure access to different sites.



Security is a moving target and what is safe and secure today may not be so tomorrow. Encryption relies on the ability for the encrypted data to be hard to crack and as computational power becomes cheaper, it is imperative that the devices providing encryption have the ability to easily implement the changes and enhancements to the TLS standard. Leveraging existing knowledge in the IT world and implementing some of their techniques to improve security is a step in the right direction.

Security is a moving target and what is safe and secure today may not be so tomorrow. Encryption relies on the ability for the encrypted data to be hard to crack and as computational power becomes cheaper, it is imperative that the devices providing encryption have the ability to easily implement the changes and enhancements to the TLS standard. Leveraging existing knowledge in the IT world and implementing some of their techniques to improve security is a step in the right direction.

ABOUT THE AUTHOR

Harpatap Parmar is a Senior Product Manager at Contemporary Controls, which designs and manufactures BACnet building controls and networking equipment. Harpatap focuses on industrial automation, information security and IP routers. He has more than 17 years of experience at Contemporary Controls developing and testing a range of networking, control and communication products.

ASHRAE SSPC 135 Meetings BACnet Proceedings

Bernhard Isler

Chairman, ASHRAE SSPC 135 BACnet Committee

SSPC 135, the BACnet committee, conducted their spring meeting in April over four days in Plantation, Florida, which was followed by plenary meetings and working groups over five days in June as part of the ASHRAE Annual Conference in Long Beach, California. Following is an overview of the core work items, including addenda and proposals, and discussions.

Semantic Information in BACnet and ASHRAE 223P

During the meeting of the AP-WG in Long Beach, the critical question was raised whether a semantic tag vocabulary for building data defined by ASHRAE, as drafted for the planned new ASHRAE Standard 223P, would ever gain significant market relevance. In particular, Project Haystack has taken an important position, and has developed significantly over the past few years. A large community of supporters has formed in and around Project Haystack, which is actively participating in the definition of the tags. The quality and stability of these definitions appear to be sufficient for the requirements presented by applications.

The discussion revealed that a single and common tag vocabulary would better serve the industry and the customers. It was agreed that investigations are needed on how the BACnet committee and ASHRAE could work with Project Haystack for achieving an ASHRAE and ISO standard of a tag vocabulary.

BACnet/IT - BACnet/SC

In the review of Addendum 135-2016bj, BACnet/IT, a number of substantial issues and ambiguities in the proposed BACnet Virtual Router Link (BVRL) were unveiled. The overhaul of the BVRL led to an approach which is not only meeting almost all of the BACnet/IT requirements, but also remains fully backward compatible with the current BACnet stack.

A new regular data link layer for the BACnet stack, so called BACnet Secure Connect, or BACnet/SC, will be defined. This new data link layer supports a virtual hub-and-spoke topology, which also allows direct connections between devices. The spokes are based on TLS secured WebSockets. In the hub, a switch function is present for the dispatching of packets among spokes as of their virtual MAC addresses, and for the distribution of broadcasts. A redundancy concept for the switch function is providing enhanced availability.

The new BACnet/IT protocol architecture and the device proxies for the translation between two stacks are not needed anymore. It was accepted that the support of the BACnet Network Layer and the segmentation in the application layer will still be required.

The BACnet/IT concepts for a centralized directory and for group communication of devices will be pursued in a more general approach and on an independent timeline, outside of the BACnet/IT addendum. The use of DNS-SD will be investigated for potential plug-and-play features of BACnet/SC.

Status of Addenda to BACnet Standard 135-2016

Some of the addenda are now ready for publication. The committee is waiting for the completion of the review of addendum 135-2016bi for publishing a packet making up protocol revision 20. In the meetings, a number of other addenda were approved for a first or future public review, expected to be open in fall.

135-2016ap	In AP-WG for second review	Application Interfaces, is on-hold at this point. The Applications Working Group is now working on defining application interfaces using semantic tags.
135-2016bb	In PS-WG for second review	Zero configuration capabilities for MS/TP devices. With this mechanism, MS/TP master devices can determine their MAC address automatically.
135-2016bd	In third review	Adds the new Staging object type to the standard. The object controls a set of binary values in other objects, based on a commanded analog value.
135-2016be	Approved for third review	New BIBBs and device profiles for the lighting domain
135-2016bi	Approved for fourth review	Introduces a standardized audit reporting and logging mechanism for auditable actions of BACnet devices and users. Only one change from the PPR3 draft is open for review this time.
135-2016bj	In IT-WG for second review	BACnet/IT. This addendum will be realigned to introduce BACnet/SC.
135-2016bk	Ready for publication	Extends the range of reserved property identifiers above 4194303
135-2016bl	Ready for publication	Clarifications in response to official interpretation requests
135-2016bm	Ready for publication	Enhances and clarifies MS/TP on some of its aspects
135-2016bn	Approved for first review	Extends and corrects some BIBBs and device profiles.
135-2016bo	In AP-WG for first review	Semantic tags concept for BACnet
135-2016bp	Ready for publication	Corrects and tweaks the BACnet RESTful Web Services in some details
135-2016bq	Approved for first review	Complements and clarifies some smaller aspects of the Access Control objects.

Status of Addenda to BACnet Test Standard 135.1-2013

The publication of finalized addenda to the test standard 135.1-2013 is in preparation.

135.1-2013p	Ready for publication	New EPICS consistency tests.
135.1-2013q	In TI-WG for first review	Amendments of tests for covering the revision of the alarming which had been introduced with protocol revision 13
135.1-2013r	Ready for publication	Tests for new properties and error codes

The committee is looking forward to meeting again in person at Georgia Tech in Atlanta, GA for its next interim meeting, being held October 23 - 26, 2017.

Public review and final versions of addenda are available at no cost from ASHRAE (<http://www.ashrae.org>) as well as on the BACnet website (<http://www.bacnet.org>). To stay up-to-date on public reviews, publications, and interim meetings, you can subscribe to the weekly ASHRAE Standards Actions electronic newsletter, or read it, at the ASHRAE website <http://www.ashrae.org/standards-research-technology/standards-actions>.

ABOUT THE AUTHOR

Bernhard Isler, Chairman ASHRAE SSPC 135 BACnet committee, works for Siemens Building Technologies as system architect at its headquarters in Zug, Switzerland. He first got involved with BACnet in 1992, when evaluating BACnet,

available as its second public review draft at that time, for application in fire detection systems. In 2011, he was the recipient of the first Swan Award, set out in honor of Bill Swan.

Bernhard completed a Professional Education in Electro Mechanics with a Swiss national certification, and holds a BA in Electrical Engineering from the University of Applied Sciences in Rapperswil, Switzerland.

Have A Question About BACnet? The BACnet Institute Has Your Answer!

By Michelle Eriquez

Education & Information Initiatives, The BACnet Institute

Have a question about BACnet that is keeping you up at night? Then you should visit The BACnet Institute (TBI) to get an answer!

Depending on your preference, the TBI online center offers two ways to access the collective knowledge of the community:

Technical Forum

This is an open discussion (threaded) forum. It encourages knowledge sharing across differing expertise and backgrounds within the BACnet community, resulting in a wide variety of perspectives in regards to innovative strategies, research, and best practices. The topics are just as diverse as the community, ranging from simplistic to highly advanced.

Ask the Expert

This tool uses a questions and answer format, providing only one answer per question submitted. The final answer is collectively composed by a panel of experts, ensuring it is as accurate and precise as possible.

Don't have a question to ask?

That's OK! The community is still an incredible source of information. Use the search functionality in the forums to find past posts on a variety of topics including network security, web services, and other topics like the functionality of BACnet Objects, Properties and Services. The information provided here is robust. In fact, past discussions submitted through the Cornell University BACnet-L email list server have also been captured in the Technical Forum. This provides even more visibility and added functionality to these discussions.

Don't be shy. [Visit TBI today](#) and submit that burning question, submerge yourself in past discussions, or even better, share your own expertise by participating in a discussion. Join us and be part of a community of BACnet professionals like yourself.

COMING SOON!!

TBI's next course is currently in its final stages of production (*projected launch date – early October*)! This new course provides valuable insight on how to effectively and efficiently implement a BACnet-based BAS in your building, all from a facility manager's perspective. This course will discuss best practices you should employ, as well as the pitfalls to avoid through the full design process – specification, procurement, construction and commissioning. If you are, or you know someone who is, starting a BACnet-based BAS project, then this course is a must. Also, like the current BACnet Basics course, this new course will be on-demand and self-paced so you can access it at anytime, anywhere. And it is free!



The BACnet Institute (TBI) is an e-learning environment developed to serve as a central source for BACnet-related education as well as a vehicle for deeper collaboration and knowledge exchange. The goal of TBI is to assert BACnet's essential role in building automation as well as ensure its successful integration. With TBI, the BACnet community, spanning development, integration, engineering and facility management professionals, can now conveniently share innovative strategies, research, and best practices from anywhere around the world, at any time of the day.

While registration to TBI is required, it is free. New users can register by selecting 'Sign Up Now' on the login page (<http://www.thebacnetinstitute.org>). Most importantly, if you have any suggestions on educational content or site functionality, we encourage you to submit these through our [online form](#). Your suggestions help us enhance The BACnet Institute e-learning environment. Thank you!



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BACnet International is an industry association that facilitates the successful use of the BACnet protocol in building automation and control systems through interoperability testing, educational programs and promotional activities.



BACnet is leading the world in Building Protocol Standard: It plays a significant role in building automation projects worldwide.



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BACnet Testing Laboratories

bacnetlabs.org

btl-coordinator@bacnetinternational.org

+1-770-971-6003



1827 Powers Ferry Road
Building 14, Suite 100
Atlanta, GA 30339
p: (770) 971-6003
f: (678) 229-2777
www.bacnetinternational.org